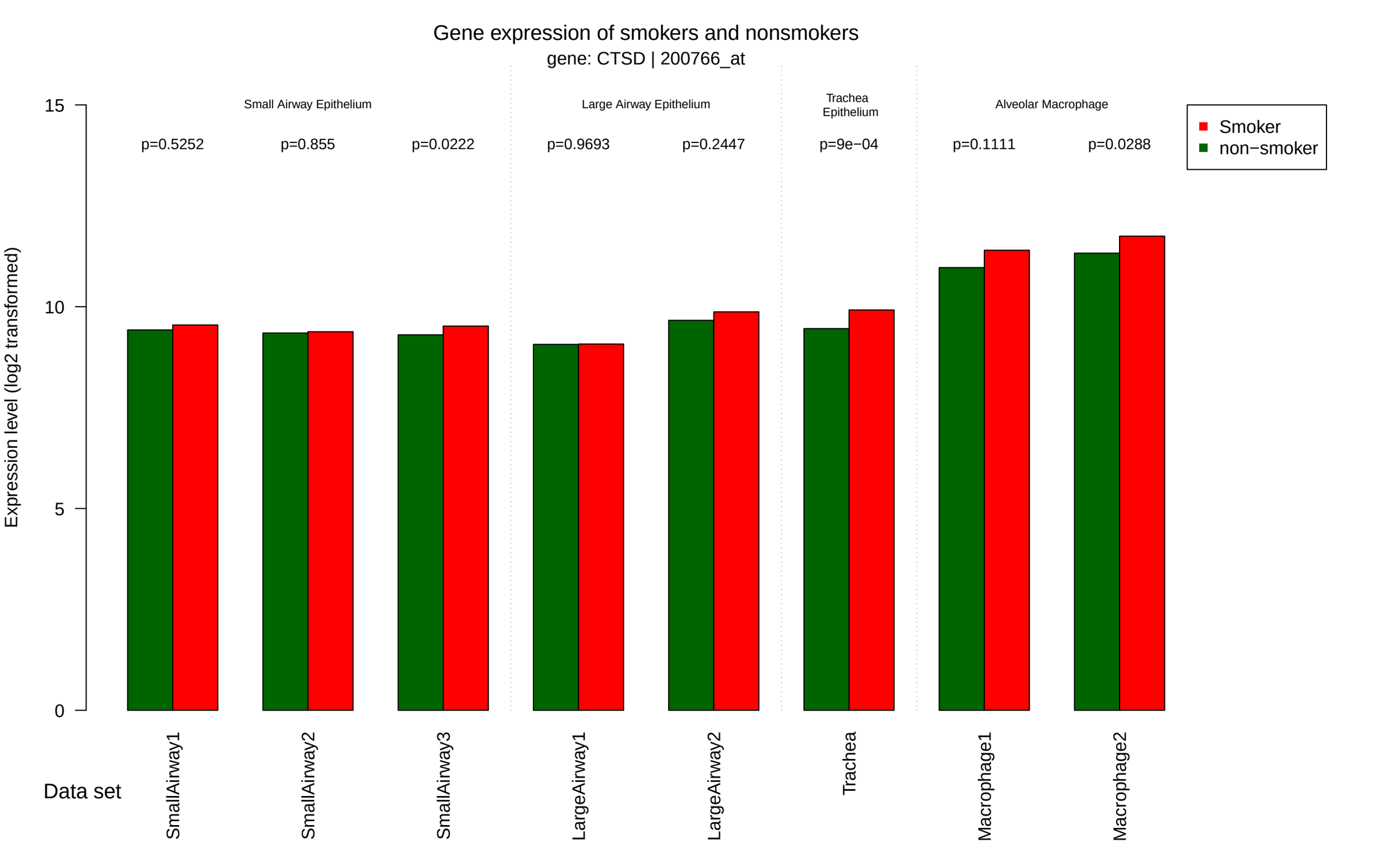
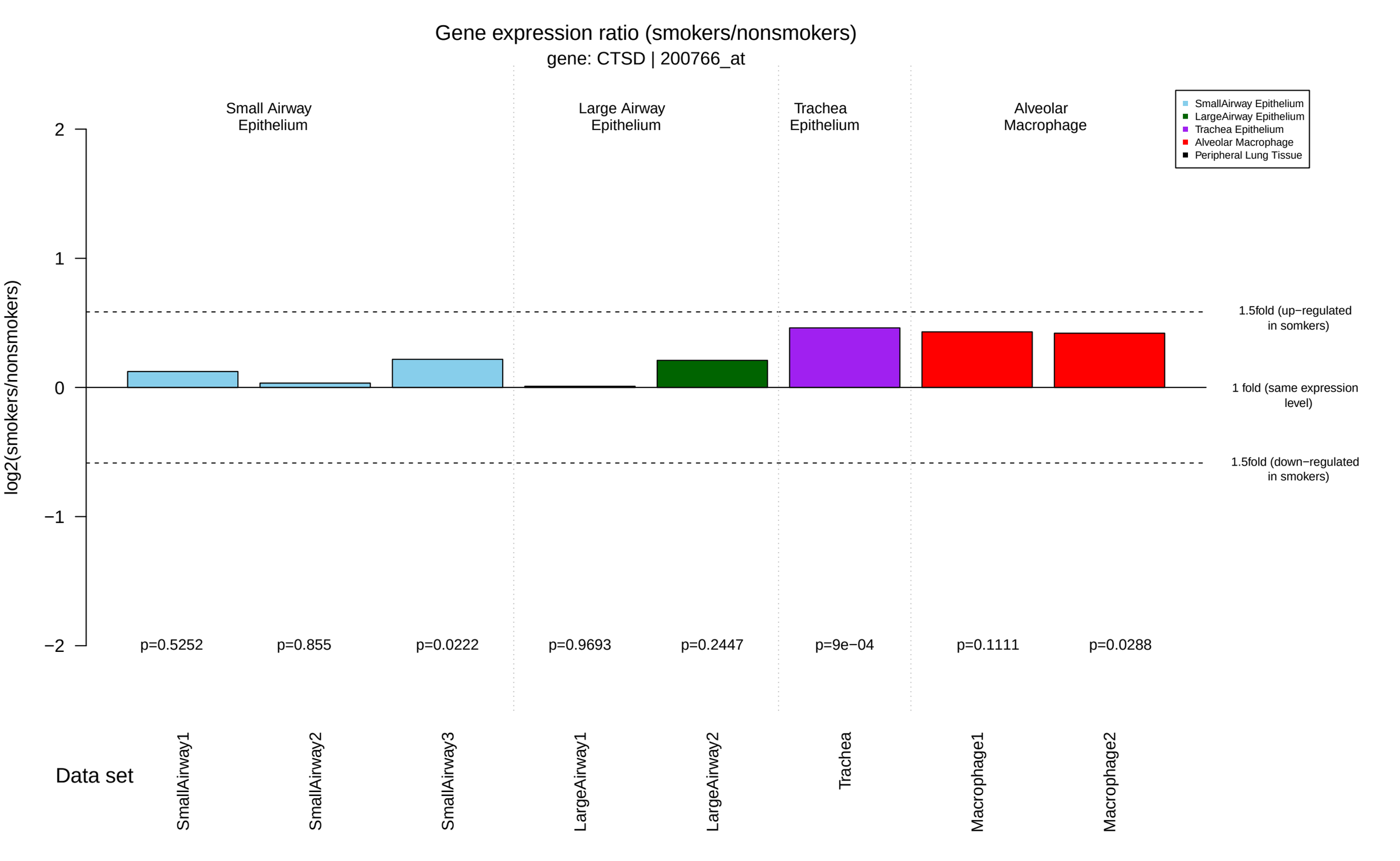
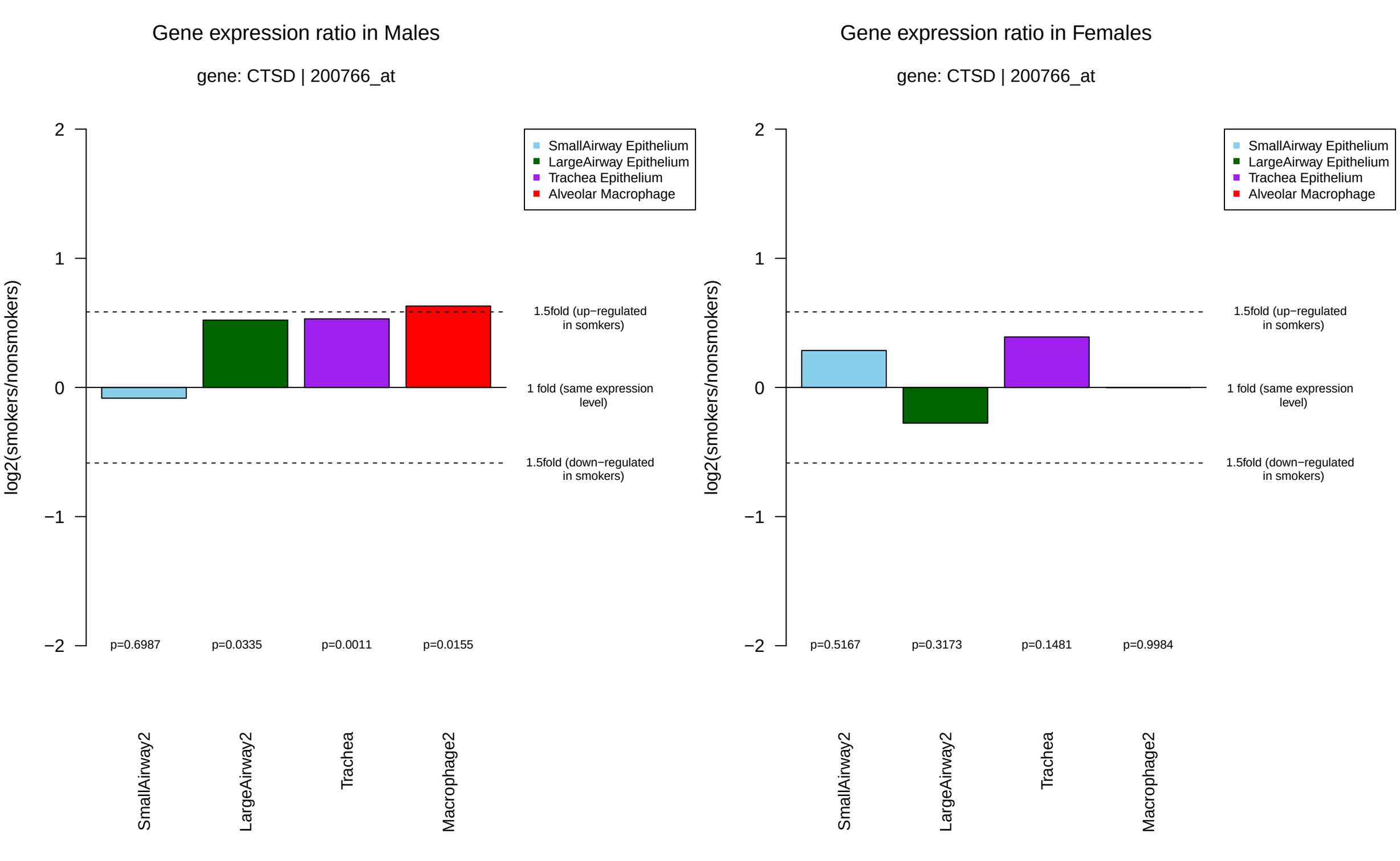
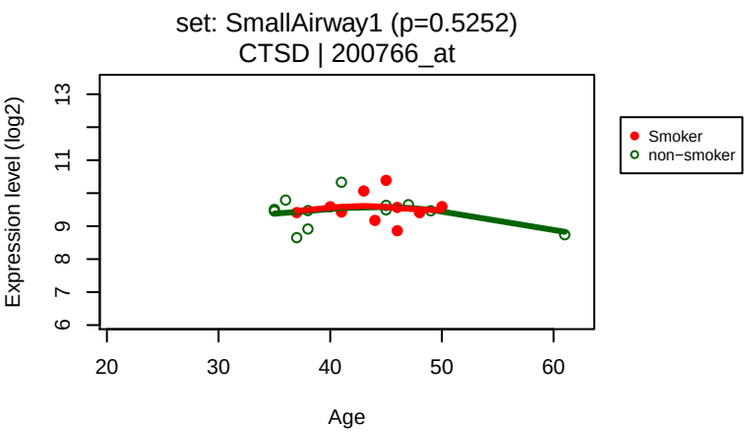
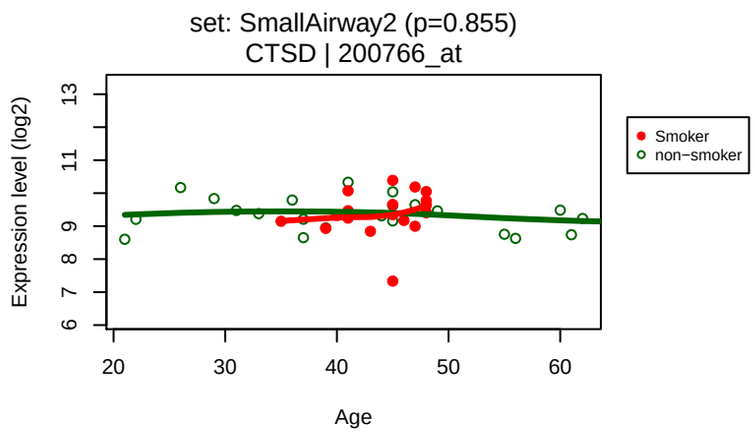
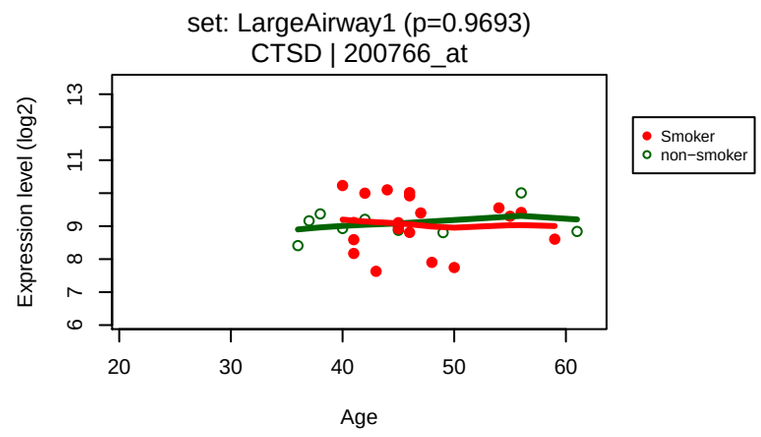

Gene expression of smokers and nonsmokers
gene: CTSD | 200766_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5252
p=0.855
p=0.0222
p=0.9693
p=0.2447
p=9e−04
p=0.1111
p=0.0288
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

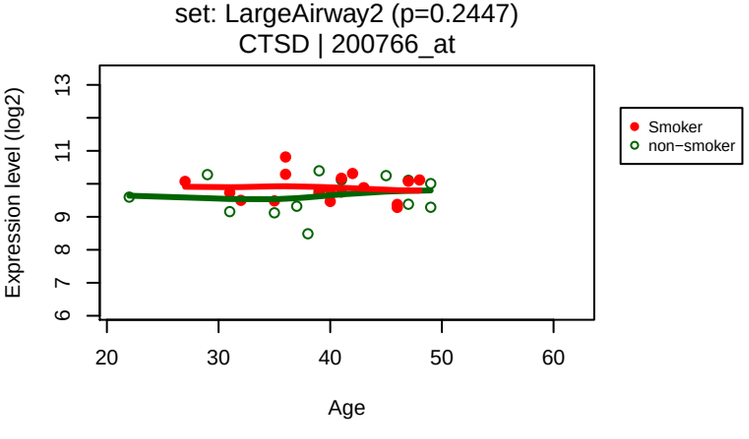

Gene expression ratio (smokers/nonsmokers)
gene: CTSD | 200766_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5252
p=0.855
p=0.0222
p=0.9693
p=0.2447
p=9e−04
p=0.1111
p=0.0288
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

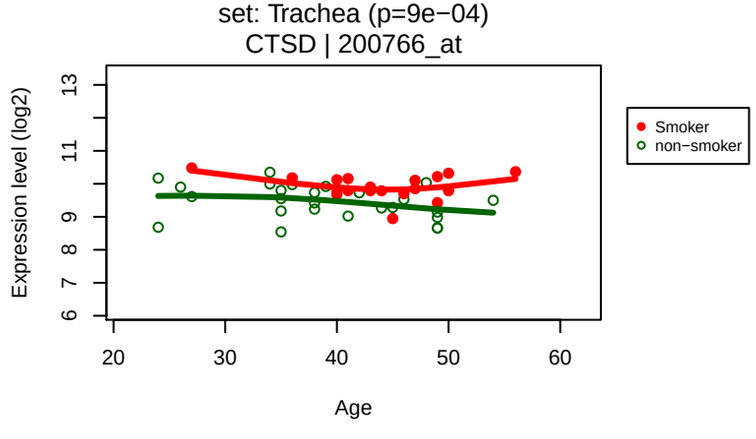

Gene expression ratio in Males
gene: CTSD | 200766_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6987
p=0.0335
p=0.0011
p=0.0155
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CTSD | 200766_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5167
p=0.3173
p=0.1481
p=0.9984
SmallAirway2
LargeAirway2
Trachea
Macrophage2

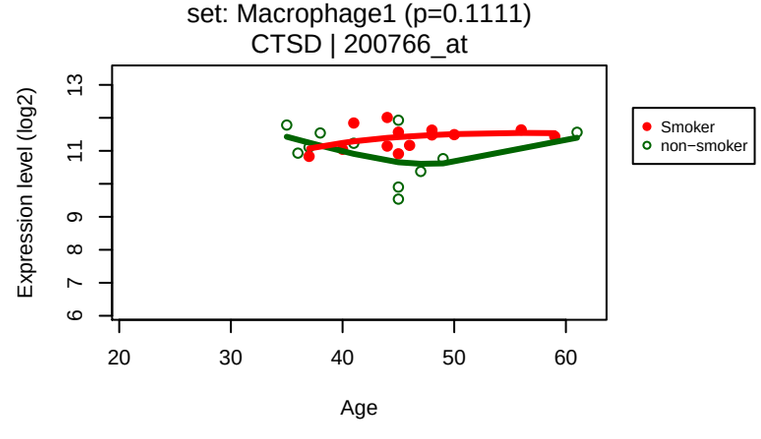

set: SmallAirway1 (p=0.5252)
CTSD | 200766_at
Expression level (log2)
Age
Smoker
non−smoker

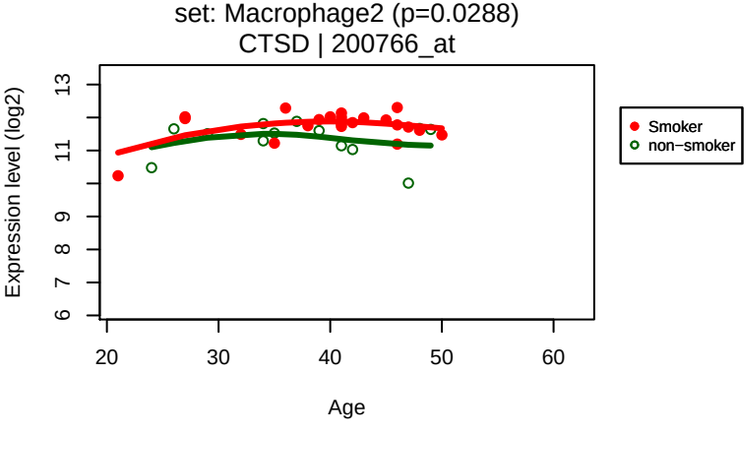

set: SmallAirway2 (p=0.855)
CTSD | 200766_at
Expression level (log2)
Age
Smoker
non−smoker

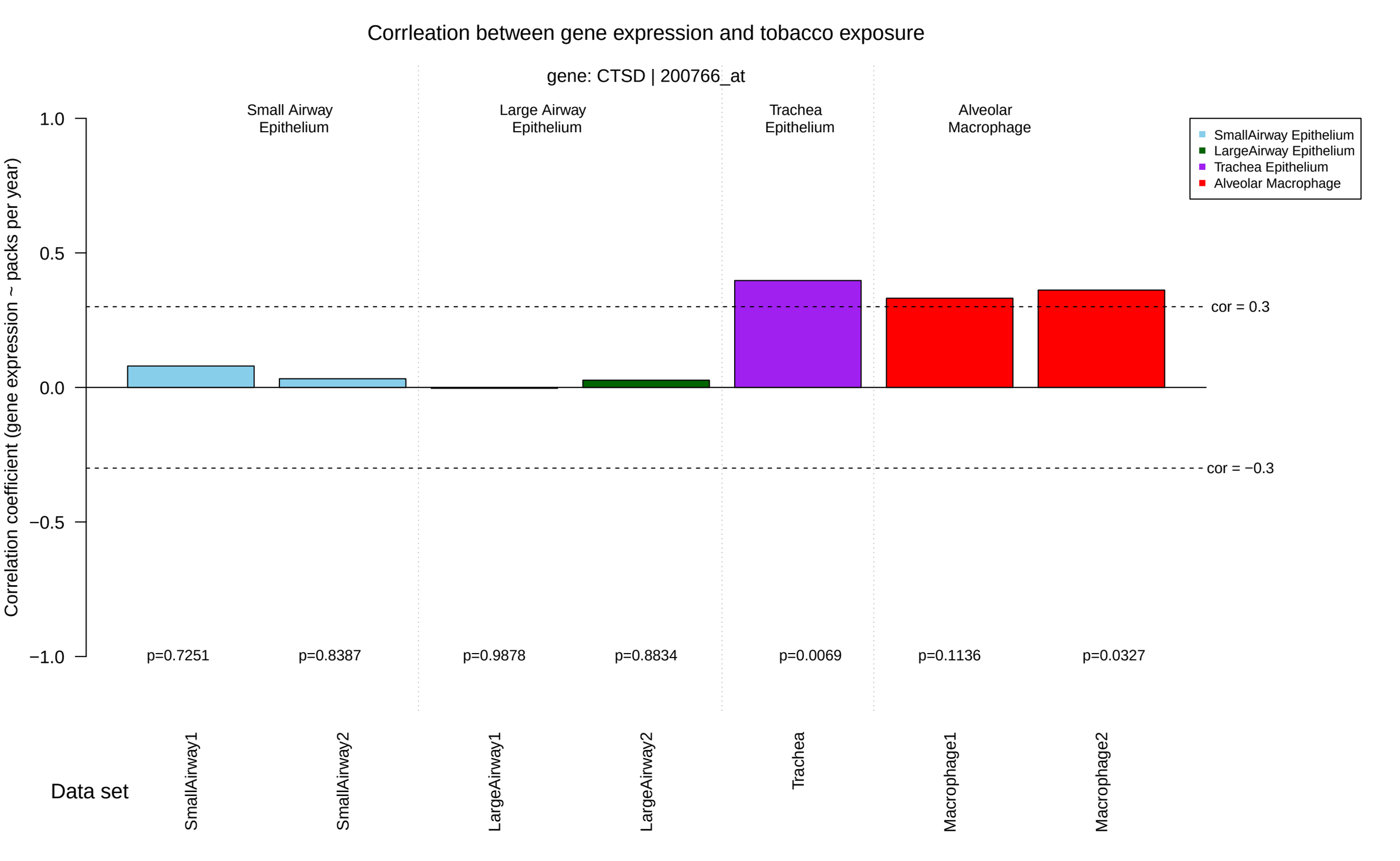

set: LargeAirway1 (p=0.9693)
CTSD | 200766_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2447)
CTSD | 200766_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=9e−04)
CTSD | 200766_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1111)
CTSD | 200766_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0288)
CTSD | 200766_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CTSD | 200766_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7251
p=0.8387
p=0.9878
p=0.8834
p=0.0069
p=0.1136
p=0.0327
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2